

STUDY GUIDE

Introduction to Endocrine Physiology

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PHS 220 — Physiology

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§1 HOMEOSTASIS & THE ENDOCRINE SYSTEM

Endocrine physiology is fundamentally concerned with the maintenance of homeostasis — the stable internal milieu that allows cells and organ systems to function optimally despite a constantly changing external environment. The endocrine system achieves this by releasing chemical messengers called **hormones** into the bloodstream, which then travel to distant target tissues to elicit specific biological responses.

The word *hormone* derives from the Greek *hormaein*, meaning 'to excite' or 'to set in motion.' Hormones are soluble factors secreted in minute quantities (picomolar to nanomolar range) that act on target cells expressing specific receptor proteins. Their effects are diverse: they regulate metabolism, growth, reproduction, fluid and electrolyte balance, and the body's responses to stress.

CORE PRINCIPLE

The endocrine system does not act in isolation — it is deeply integrated with the nervous and immune systems. Neuroendocrine interactions (e.g., the hypothalamo-pituitary axis) and immunoendocrine cross-talk (e.g., cortisol suppressing inflammation) are central to understanding physiology and disease.

Types of Chemical Signalling

Signalling Type	Route	Distance	Example
Endocrine	Bloodstream	Long-range (systemic)	Insulin from pancreas → liver
Paracrine	Interstitial fluid	Local (neighbouring cells)	Somatostatin from delta cells → beta cells
Autocrine	Same cell	Cell acts on itself	Growth factors in tumour cells
Neurocrine	Synapse / neuroendocrine	Targeted via axon	ADH from hypothalamus → kidney
Juxtacrine	Direct cell contact	Adjacent cells only	Notch-Delta signalling

MAJOR HORMONAL CONTRIBUTORS TO HOMEOSTASIS

Thyroid Hormone — controls basal metabolism in most tissues.

Cortisol — energy metabolism; permissive action for other hormones.

Aldosterone / Mineralocorticoids — regulate plasma volume via electrolytes.

Vasopressin (ADH) — regulates plasma osmolality via water excretion.

PTH — regulates calcium & phosphate levels.

Insulin — regulates plasma glucose concentration.

§2 CLASSES OF HORMONES

Hormones are chemically diverse. Their chemical nature determines how they are synthesised, stored, transported, and how they signal to target cells. There are three principal classes:

2.1 Peptide / Protein Hormones

Peptide hormones are the most numerous class and range from small peptides (e.g., TRH — 3 amino acids) to large glycoproteins (e.g., FSH, TSH). Key characteristics:

- Synthesised on ribosomes as pre-prohormones → prohormones → active hormones (e.g., insulin from proinsulin).
- Water-soluble — travel freely in plasma; short half-life (minutes).
- Cannot cross the cell membrane — act on **cell-surface receptors** (GPCRs, receptor tyrosine kinases, cytokine receptors).
- Signal via second messengers: cAMP, IP₃/DAG, Ca²⁺.
- Examples: Insulin, Glucagon, GH, ACTH, PTH, GnRH, LH, FSH, Oxytocin, ADH.

2.2 Steroid Hormones

Steroid hormones are derived from cholesterol through a series of enzymatic conversions that occur primarily in the adrenal cortex, gonads, and placenta. Key characteristics:

- Lipid-soluble — cross plasma membranes freely; long half-life (hours).
- Travel in plasma bound to carrier proteins (e.g., CBG, SHBG, albumin); only unbound (free) fraction is active.
- Act on **intracellular (nuclear) receptors** — hormone-receptor complex binds to hormone response elements (HREs) on DNA to regulate gene transcription.
- Effects are slow (hours) but long-lasting.
- Examples: Cortisol, Aldosterone, Testosterone, Oestrogen, Progesterone, Vitamin D (calcitriol).

2.3 Amine (Tyrosine-Derived) Hormones

Amine hormones are derived from the amino acid tyrosine and exhibit hybrid properties that span the peptide and steroid categories:

Hormone	Gland	Solubility	Receptor Type	Half-life
Adrenaline / Noradrenaline	Adrenal medulla	Water-soluble	Cell surface (α/β-AR)	Seconds-minutes
Dopamine	Hypothalamus/adrenal	Water-soluble	Cell surface (D-receptors)	Minutes
Thyroid hormones (T3/T4)	Thyroid gland	Lipid-soluble	Nuclear (TR)	T4: ~7 days; T3: ~1 day

§3 HORMONE RECEPTORS

A receptor is a protein that specifically binds a hormone (the ligand) with high affinity and selectivity, and upon binding, triggers a biological response in the target cell. The identity and number of receptors expressed on a cell determine its sensitivity to a particular hormone.

3.1 Cell-Surface (Membrane) Receptors

Receptor Class	Coupling Mechanism	Second Messenger	Hormone Examples
G-protein coupled receptors (GPCRs) — Gs	Adenylyl cyclase activation ↑	cAMP → PKA	Glucagon, TSH, ACTH, LH, FSH, PTH, Adrenaline (β)
GPCRs — Gi	Adenylyl cyclase inhibition ↓	↓ cAMP	Somatostatin, Adrenaline (α2)
GPCRs — Gq	Phospholipase C activation	IP3 + DAG → Ca ²⁺ , PKC	GnRH, TRH, Oxytocin, Adrenaline (α1)
Receptor Tyrosine Kinases (RTKs)	Autophosphorylation	RAS-MAP kinase, PI3K-Akt	Insulin, IGF-1, EGF, PDGF
JAK-STAT receptors	JAK activation	STAT phosphorylation	Growth hormone, Prolactin, Cytokines
Receptor Guanylyl Cyclases	Direct GC activation	cGMP → PKG	ANP, BNP

3.2 Intracellular Receptors

Intracellular receptors are used by lipophilic hormones that can cross the plasma membrane. These receptors function as ligand-activated transcription factors. Upon hormone binding, they dimerize and translocate to the nucleus, where they bind hormone response elements (HREs) in the promoter regions of target genes, initiating or repressing transcription.

RECEPTOR REGULATION

Down-regulation: Prolonged exposure to a hormone leads to a decrease in receptor number (internalisation/degradation). This is the basis of tachyphylaxis — reduced response despite continuing stimulation (e.g., prolonged GnRH agonist therapy paradoxically suppresses LH/FSH).

Up-regulation: Low hormone levels or opposing hormones can increase receptor number, increasing target cell sensitivity (e.g., oestrogen up-regulates progesterone receptors in the uterus).

§4 PRINCIPLES OF FEEDBACK CONTROL

Feedback control loops are the primary mechanism by which the endocrine system maintains hormonal concentrations within a narrow physiological range. The hypothalamo-pituitary-target organ axes exemplify multi-level feedback regulation.

4.1 Negative Feedback (Most Common)

In negative feedback, the product of a hormonal cascade feeds back to inhibit upstream components, preventing excessive hormone action. This is the dominant regulatory mechanism in the HPT (hypothalamo-pituitary-thyroid), HPA (hypothalamo-pituitary-adrenal), and HPG (hypothalamo-pituitary-gonadal) axes.

Example — HPA axis: CRH (hypothalamus) → ACTH (anterior pituitary) → Cortisol (adrenal cortex). Elevated cortisol feeds back to suppress both CRH and ACTH secretion, preventing further cortisol rise.

4.2 Positive Feedback

In positive feedback, the product amplifies the upstream signal, driving a cascade to completion. This is less common but physiologically critical at specific moments. The most important endocrine example is the mid-cycle LH surge: rising oestrogen — once above a threshold — switches from negative to positive feedback on the pituitary, triggering the explosive LH surge that causes ovulation.

§5 PULSATILE SECRETION & CIRCADIAN RHYTHMS

Many hormones are not secreted continuously but in discrete pulses. Pulsatile secretion serves several key functions: it prevents receptor down-regulation, allows for precise amplitude modulation, and creates windows of sensitivity in target tissues. The master circadian pacemaker is located in the **suprachiasmatic nucleus (SCN)** of the hypothalamus, which is entrained by light-dark cycles and drives rhythmic hormone secretion.

Hormone	Rhythm Pattern	Peak	Clinical Relevance
Cortisol	Circadian (~24 hr)	Early morning (6–8 AM)	Loss of rhythm in Cushing syndrome; measure AM cortisol
GH	Pulsatile; nocturnal	Shortly after sleep onset	Continuous GH secretion has reduced anabolic effect
LH/FSH	Pulsatile (GnRH-driven)	Vary across menstrual cycle	Continuous GnRH → receptor down-regulation → suppression
TSH	Circadian (mild)	Late evening/night	Blunted pulse in hypothyroidism
Melatonin	Circadian	During darkness	Regulates sleep-wake and seasonal reproduction
Insulin	Prandial + basal pulses	Post-meal	Pulsatile delivery to liver more effective than continuous

§6 ENDOCRINE ORGANS

6.1 Traditional Endocrine Organs

Organ	Location	Key Hormones	Primary Function
Hypothalamus	Floor of diencephalon	TRH, CRH, GnRH, GHRH, Somatostatin, Dopamine	Master regulator; integrates neural & endocrine signals
Anterior Pituitary	Sella turcica (sphenoid)	TSH, ACTH, GH, FSH, LH, Prolactin	Trophic hormones governing peripheral glands
Posterior Pituitary	Sella turcica	ADH (Vasopressin), Oxytocin	Released from hypothalamic axon terminals
Thyroid Gland	Anterior neck	T3, T4, Calcitonin	Basal metabolic rate; calcium regulation
Parathyroid Glands (4)	Posterior thyroid	PTH	Calcium & phosphate homeostasis
Adrenal Cortex	Retroperitoneal (supra-renal)	Cortisol, Aldosterone, DHEA	Stress response; fluid balance; sex hormones
Adrenal Medulla	Core of adrenal gland	Adrenaline, Noradrenaline	Fight-or-flight response (sympathetic)
Pancreatic Islets	Scattered in exocrine pancreas	Insulin (β), Glucagon (α), Somatostatin (δ)	Glucose homeostasis
Gonads (Ovary/Testis)	Pelvis / Scrotum	Oestrogen, Progesterone / Testosterone	Reproduction; secondary sex characteristics
Pineal Gland	Posterior diencephalon	Melatonin	Circadian rhythm; seasonal reproduction

6.2 Non-Traditional Endocrine Organs

Many organs not classically considered endocrine glands nonetheless secrete hormones with important systemic effects:

Organ	Hormone(s)	Function
Heart (atria)	ANP, BNP	Natriuresis; reduce BP and blood volume
Kidney	Erythropoietin (EPO), Renin, Calcitriol (1,25-OH-D3)	RBC production; RAAS; calcium absorption
Liver	IGF-1, Angiotensinogen, Thrombopoietin	Growth; blood pressure regulation; platelet production
Adipose Tissue	Leptin, Adiponectin, Resistin	Satiety; insulin sensitivity; energy balance
GI Tract	GIP, GLP-1, CCK, Ghrelin, Secretin	Digestion; incretin effect; appetite regulation
Skin	Cholecalciferol (Vit D precursor)	UV-driven Vit D synthesis
Placenta	hCG, hPL, Oestrogen, Progesterone	Pregnancy maintenance; fetal development
Skeletal Muscle	Myokines (Irisin, IL-6)	Exercise-induced metabolic adaptations

§7 THE HYPOTHALAMO-PITUITARY AXIS

The hypothalamo-pituitary axis represents the supreme command centre of the endocrine system. The hypothalamus receives input from higher cortical centres, limbic system, and peripheral hormonal signals, and translates this into a coordinated hormonal output via the pituitary gland.

7.1 The Hypothalamus

The hypothalamus secretes releasing and inhibiting hormones into the **hypothalamo-hypophyseal portal system** — a specialised vascular link between the hypothalamus and anterior pituitary. This portal system allows hormones secreted in minute amounts to reach the pituitary at concentrations 1,000-fold higher than in the systemic circulation.

Hypothalamic Hormone	Action on Anterior Pituitary
TRH (Thyrotropin-Releasing Hormone)	Stimulates TSH and Prolactin release
CRH (Corticotropin-Releasing Hormone)	Stimulates ACTH release
GnRH (Gonadotropin-Releasing Hormone)	Stimulates LH and FSH release (pulsatile required)
GHRH (Growth Hormone-Releasing Hormone)	Stimulates GH release
Somatostatin (GHIH)	Inhibits GH and TSH release
Dopamine (PIH — Prolactin Inhibiting Hormone)	Inhibits Prolactin release (tonic inhibition)

7.2 Anterior Pituitary (Adenohypophysis)

Cell Type	Hormone	Target / Effect
Somatotrophs (50%)	GH (Growth Hormone / Somatotropin)	Liver → IGF-1 → growth, lipolysis, gluconeogenesis
Corticotrophs (20%)	ACTH (Adrenocorticotrophic Hormone)	Adrenal cortex → cortisol, androgens
Thyrotrophs (5%)	TSH (Thyroid-Stimulating Hormone)	Thyroid → T3/T4 production and release
Gonadotrophs (10%)	FSH & LH (Gonadotropins)	Gonads → sex steroids, gametogenesis, ovulation
Lactotrophs (15%)	Prolactin	Breast → milk production; inhibits GnRH axis

7.3 Posterior Pituitary (Neurohypophysis)

The posterior pituitary does not produce hormones itself. It stores and releases hormones (ADH and Oxytocin) synthesised in the supraoptic and paraventricular nuclei of the hypothalamus, which travel down axons to be released into the systemic circulation.

Hormone	Synthesis Site	Main Stimuli for Release	Key Actions
ADH (Vasopressin)	Supraoptic nucleus	Hyperosmolality; hypovolaemia; pain	V2 receptors → water reabsorption in collecting duct; V1 receptors → vasoconstriction
Oxytocin	Paraventricular nucleus	Cervical dilation (Ferguson reflex); suckling	Uterine contraction; milk let-down; social bonding

§8 PRINCIPLES GOVERNING DISEASE STATES

Endocrine diseases can be broadly categorised into conditions of hormone excess, hormone deficiency, or resistance to hormone action. Understanding the axis affected and the level of the lesion is essential for diagnosis and management.

Disease Category	Definition	Common Cause	Lab Pattern	Example
Primary hypo-function	Failure of the target gland	Autoimmune, infarction, surgical removal	Low end-hormone; HIGH trophic hormone (loss of negative feedback)	Addison's disease (primary adrenal failure) — low cortisol, HIGH ACTH
Secondary hypo-function	Failure of pituitary (or hypothalamus)	Pituitary tumour/ infarction, cranial irradiation	Low end-hormone; Low trophic hormone	Secondary hypothyroidism — low T4, low TSH
Primary hyper-function	Autonomous over-secretion by target gland	Adenoma, hyperplasia, carcinoma	HIGH end-hormone; LOW trophic hormone (suppressed by feedback)	Conn's syndrome (primary hyperaldosteronism) — high aldosterone, low renin
Secondary hyper-function	Excess trophic stimulation	Pituitary adenoma, ectopic ACTH	HIGH end-hormone; HIGH trophic hormone	Cushing's disease — high cortisol, high ACTH (pituitary adenoma)
Hormone resistance	Target tissue unresponsive to hormone	Receptor mutation, signalling defect	HIGH hormone level; end-organ features of deficiency	Type 2 DM (insulin resistance) — high insulin, hyperglycaemia
Ectopic secretion	Non-endocrine tumour secretes hormone	Small cell lung cancer (ACTH)	HIGH hormone; PRIMARY source not gland	Ectopic ACTH syndrome — ACTH from lung tumour

CLINICAL PEARL

The pattern of trophic hormone vs. end-hormone is the key to localising the level of lesion in any endocrine axis. **Primary failure → elevated trophic hormone** (loss of negative feedback). **Secondary/tertiary failure → low trophic hormone**. This principle applies equally to the HPT, HPA, and HPG axes.